

1 MOTIVATION & GAP

- Machine unlearning aims to remove the influence of selected training data from a trained model.
- Deleting the source data is not enough: its influence may remain encoded in model parameters.
- Most prior evaluations focus on image classification, leaving audio and video less explored.

The design challenge is to balance forgetting, retained utility, privacy, and computational cost.

2 RESEARCH GOAL & CONTRIBUTIONS

GOAL
Efficiently forget a target class while preserving retained knowledge — and test whether the same logit-space mechanism transfers beyond images.

- Hybrid framework** — sharded isolation + masked distillation.
- Benchmark** — 9 unlearning methods × 4 image datasets; utility, privacy, and efficiency.
- Cross-modal evaluation** — class-sweep across 6 datasets spanning image, audio, and video.

Three contributions, one objective: effective forgetting without sacrificing retained utility.

3 RELATED WORK

EXACT

- Retraining from Scratch**
Gold-standard reference, but computationally expensive.
- SISA [1]**
Isolated shards; retrain only affected shards.

APPROXIMATE

- DELETE [2]**
Masked knowledge distillation.
- Amnesiac**
Reverses target-sample training updates.
- Boundary Unlearning**
Moves the decision boundary away from the target class.

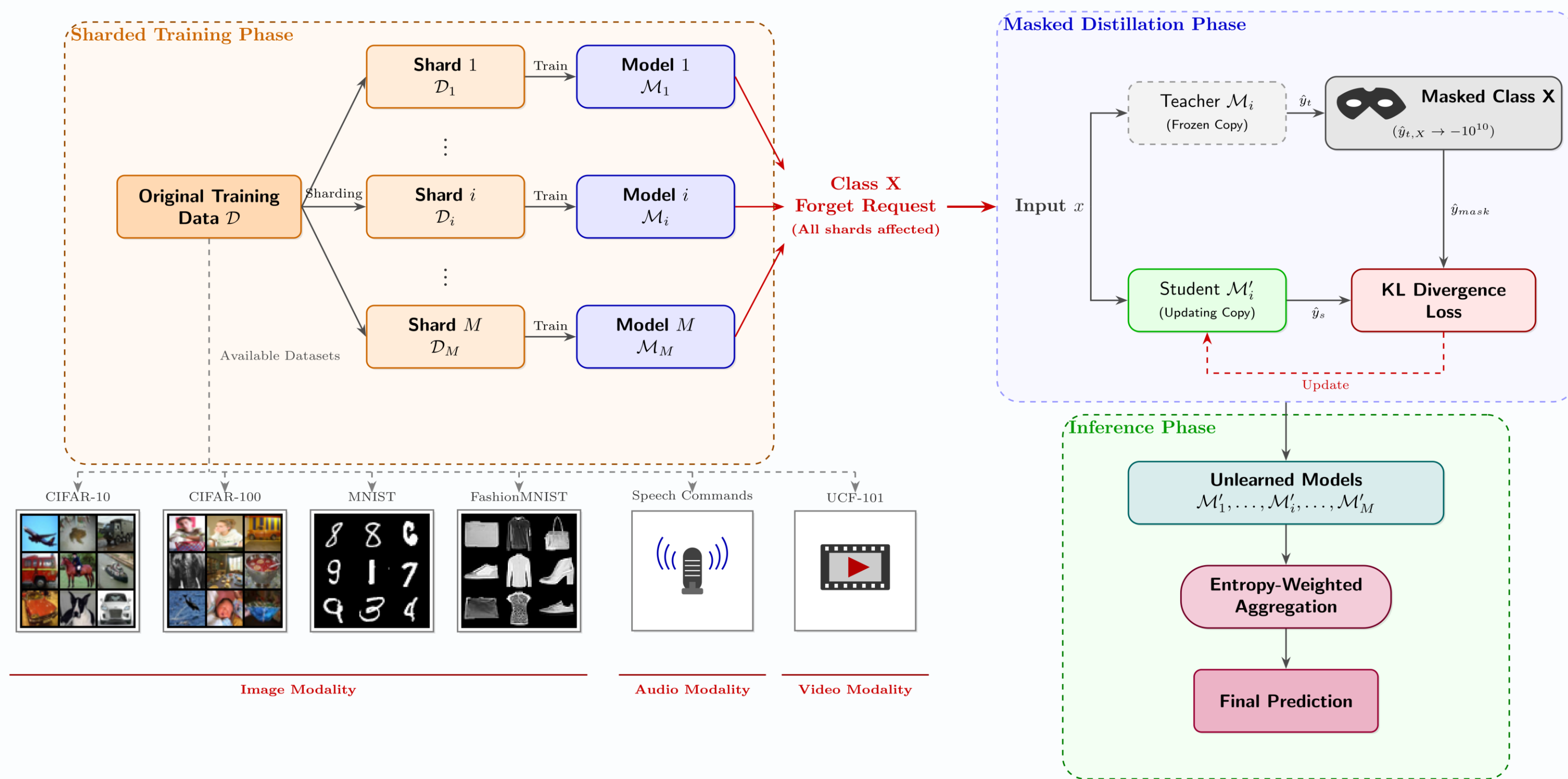
MULTIMODAL GAP

- Image-centric literature**
Most prior studies focus on image-based datasets.
- Audio & video**
Remain comparatively untouched in machine unlearning.

MIST bridges these threads: SISA-style isolation + DELETE-inspired masking, then evaluates transfer across modalities.

4 PROPOSED METHOD: MIST

MIST system overview



How the Technique Works?

1 Sharded Training

Split data into independent shards and train one model per shard. Indexes are shuffled with a fixed seed before splitting, so each shard stays class-balanced. Structural isolation keeps unlearning local to the shard being repaired.

2 Masked Distillation

Freeze the teacher, mask the forget-class logit to a very large negative value, and train a student to match it with KL divergence. The mask pushes probability mass onto the remaining classes, while the weight λ balances how hard the model forgets against how much retained knowledge it keeps.

3 Entropy-Weighted Aggregation

Unlearning damages some shards more than others. At inference each shard is weighted by the entropy of its prediction, so the more confident, less damaged shards dominate the final decision.

Optimization & Scope

Regularization Cross Entropy Loss on the retain side + gradient clipping

λ balances forgetting against retention; γ caps the gradient norm so forget batches cannot destabilise the retained weights.

5 EXPERIMENTAL SETUP & MULTIMODAL EXPANSION

Cross-modal Expansion

Image Image classification · CIFAR-10 · CIFAR-100 · MNIST · FashionMNIST
ResNet-50 backbone in the class-sweep phase

Audio Audio keyword spotting · Speech Commands
AudioCNN backbone + retain-only repair after masked distillation

Video Video action recognition · UCF101
3D ResNet-18 backbone for spatiotemporal video unlearning

Across modalities, the same logit-masking framework is preserved; only the backbone, preprocessing, and modality-specific hyperparameters change.

Method-level Image Benchmark

Compared against 9 baselines on 4 image datasets:
Retrain from Scratch · SISA · DELETE · Fine-Tuning · Gradient Ascent · SCRUB · AmnesiacML · Boundary Shrink · Boundary Expanding

Evaluation Metrics

- Forget accuracy
- Retain accuracy / F1
- RAP retention ratio
- H-Mean balance
- MIA privacy
- Unlearning time

Fair-comparison protocol: forget class = class 4 for the image benchmark; all methods share the same dataset split and evaluation procedure.

6 MAIN BENCHMARK RESULTS

0.00% test forget accuracy on all four image benchmarks

99.54 highest MIST H-Mean (MNIST)

156-300 s MIST unlearning time across image benchmarks

Method-level Comparison on Image Benchmarks

Dataset	Method	Acc _f ↓	Acc _r ↑	RAP _{te} ↑	H-Mean ↑	MIA ↓	Time (s) ↓
CIFAR-10	Retrain from Scratch	0.00%	94.62%	99.89%	97.24	0.00%	3925.2
	SISA [1]	0.00%	93.24%	98.19%	96.50	0.00%	9405.0
	DELETE [2]	0.00%	94.98%	100.27%	97.42	0.00%	626.6
	MIST (Ours)	0.00%	95.18%	100.22%	97.53	0.00%	160.1
MNIST	Retrain from Scratch	0.00%	99.27%	100.17%	99.63	0.00%	1456.5
	SISA [1]	0.00%	99.27%	100.10%	99.63	0.00%	1121.8
	DELETE [2]	0.00%	99.16%	100.06%	99.58	0.00%	583.8
	MIST (Ours)	0.00%	99.08%	99.91%	99.54	0.00%	182.2
CIFAR-100	Retrain from Scratch	0.00%	79.27%	95.54%	88.44	0.00%	4390.4
	SISA [1]	0.00%	80.36%	99.86%	89.11	0.00%	3514.4
	DELETE [2]	1.00%	79.88%	96.27%	88.42	0.87%	2456.3
	MIST (Ours)	0.00%	79.66%	98.98%	88.68	0.67%	300.2
FashionMNIST	Retrain from Scratch	0.00%	92.11%	96.62%	95.89	0.00%	1957.7
	SISA [1]	0.00%	94.96%	100.87%	97.41	0.00%	2162.5
	DELETE [2]	0.00%	95.66%	100.34%	97.78	0.00%	831.6
	MIST (Ours)	0.00%	94.89%	100.80%	97.38	0.00%	156.4

Test split, forget class = 4. Retain-side baseline for RAP: pretrained model (sharded pretrained for SISA and MIST). Full 9-baseline comparison in the paper.

Multimodal Expansion: Class-sweep H-Mean

CIFAR-10 Image		92.37	Acc _f 1.33 ± 1.24
MNIST Image		98.46	Acc _f 0.01 ± 0.03
CIFAR-100 Image		89.99	Acc _f 0.62 ± 1.21
FashionMNIST Image		97.21	Acc _f 0.00 ± 0.00
Speech Commands Audio		98.04	Acc _f 1.49 ± 1.28
UCF101 Video		89.84	Acc _f 3.39 ± 10.38

7 PRIVACY & VISUALIZATION

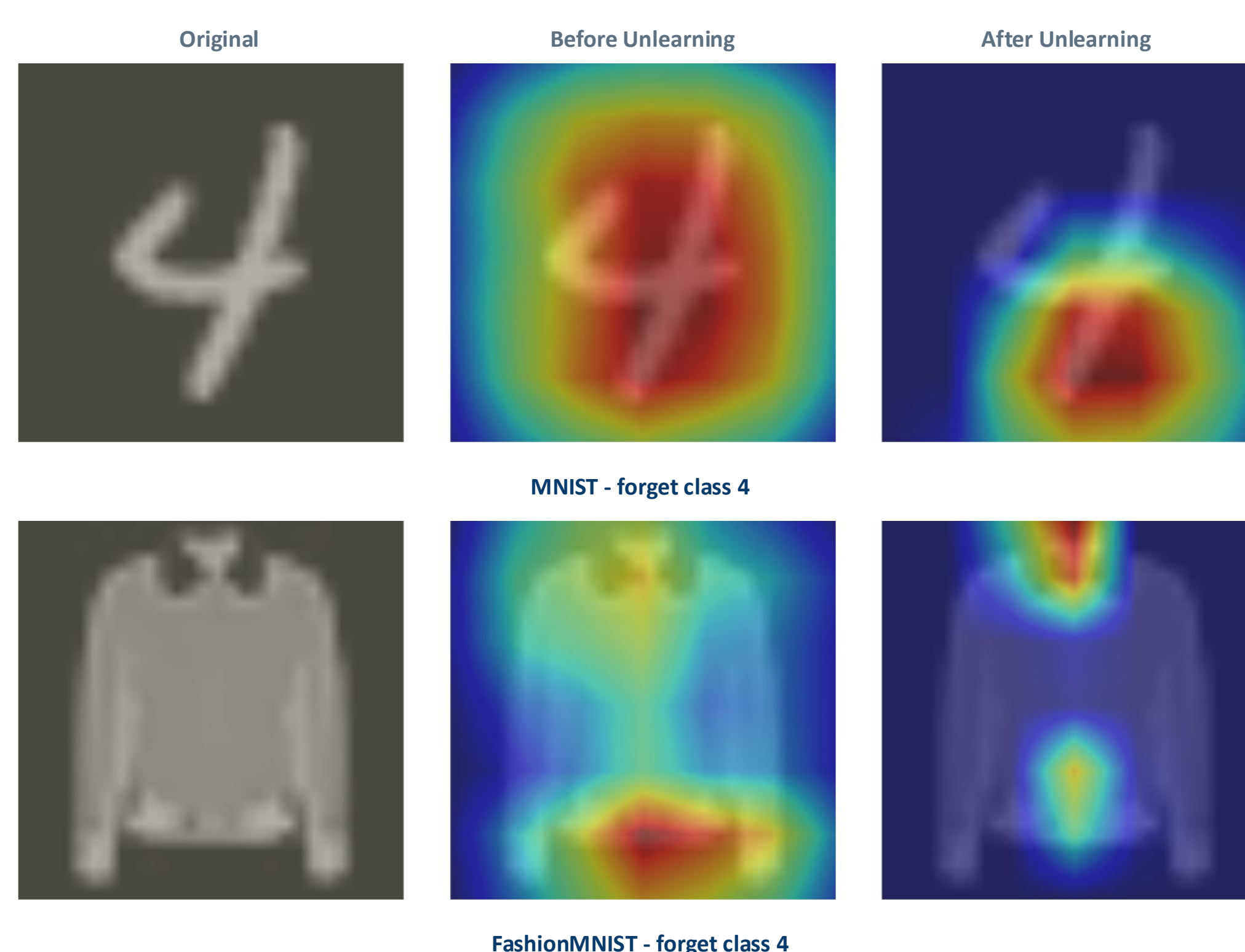
Membership Inference Attack (MIA) [3]

0.00% MIA

on both MNIST and FashionMNIST

Threshold-based MIA scored on three signals: correctness, confidence and modified entropy. A lower value means the attacker recovers less membership information about the forget samples. MIA alone does not prove complete forgetting.

Grad-CAM Attention Maps [4]



Before unlearning the model attends to the regions that identify the forget class; after unlearning the maps vanish or shift to unrelated features.

Limitation

UCF101 carries the highest class-sweep variation (Acc_f 3.39 ± 10.38, H-Mean down to 47.19 on the hardest class). Temporal motion and visual similarity between action classes make some of them far harder to isolate, so video unlearning needs per-class hyperparameter tuning.

8 CONCLUSION

Hybrid idea works

MIST combines sharded isolation with masked distillation for class-level unlearning. To the best of our knowledge it is the first unlearning method handling image, audio and video within one framework.

Evaluate more than forgetting

Strong unlearning must be judged together with retained utility, privacy and cost. Alongside H-Mean and MIA we report RAP, a retain-accuracy preservation ratio defined in this work.

Beyond images

MIST extends to audio keyword spotting (Speech Commands) and video action recognition (UCF101). Every class was swept: 10 each for CIFAR-10, MNIST, FashionMNIST and Speech Commands, 100 for CIFAR-100, 101 for UCF101.

Future work: larger multimedia datasets, improved aggregation, and sample-based or feature-based unlearning scenarios.

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- Zhou et al., "Decoupled Distillation to Erase," CVPR, 2025.
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